

PLABABLE

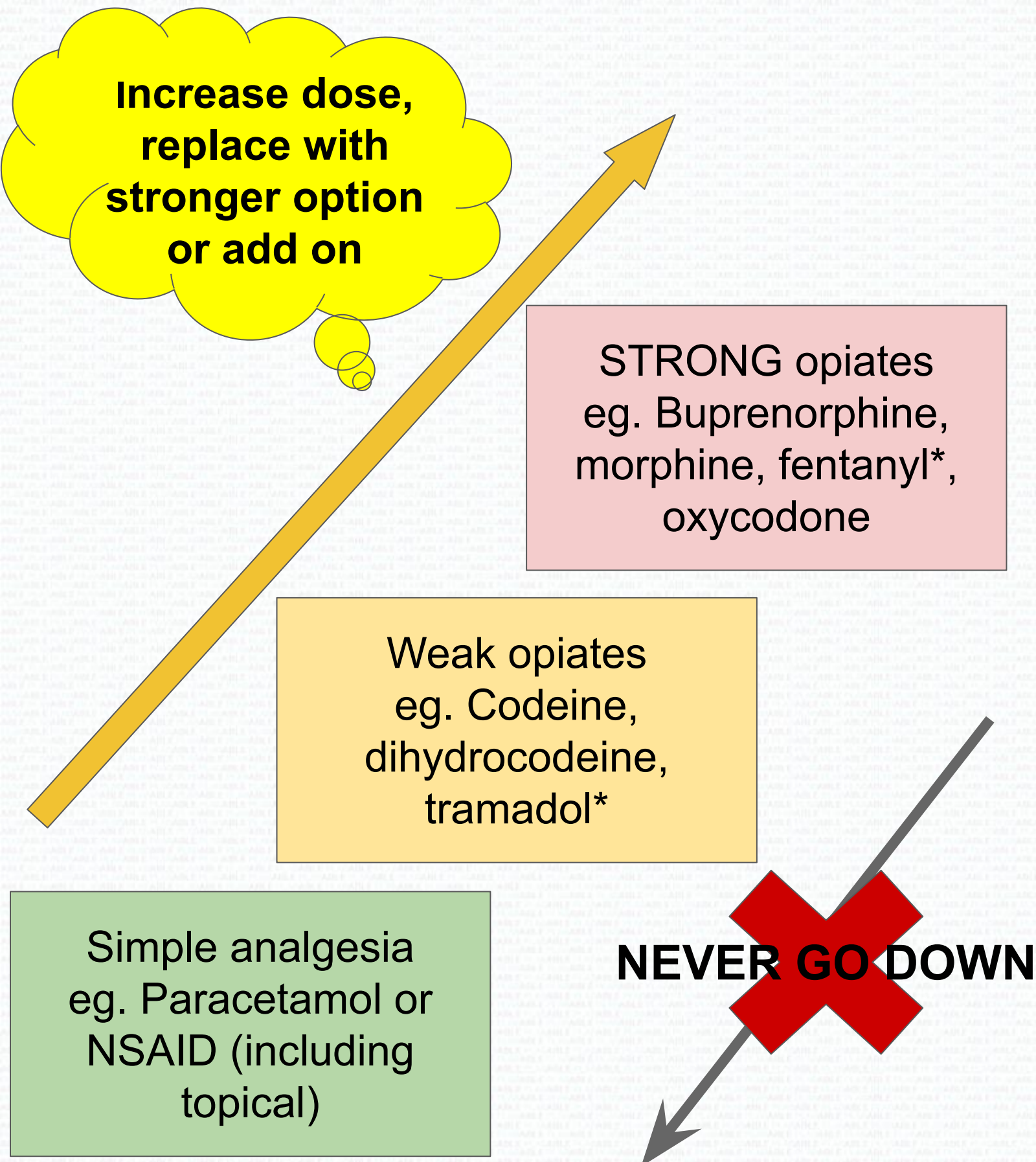
GEMS 

VERSION 1.3

PHARMACOLOGY



Pain Ladder



1. Tramadol has limited efficacy in chronic pain
2. Fentanyl patches has slow onset **NOT** suitable for acute pain

Alternative Pain Treatments

DR. A B C

Diazepam



Muscle spasm
(Short term)

Radiotherapy

- *Bisphosphonate*
(*Alendronate /*
Risedronate)



Bone metastasis

Amitriptyline /
Gabapentin / Duloxetine
/ Pregabalin



Neuropathic pain
(Shooting,
electric-shock
like, paresthesia,
burning)

Baclofen

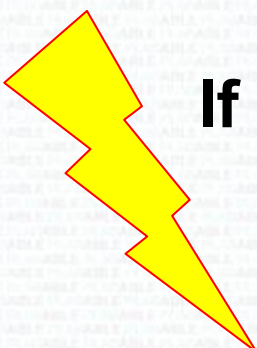


Muscle spasm
(Chronic)

Carbamazepine



Trigeminal
neuralgia



If pain is not controlled, consider
increase dose or add on
medications from pain ladder



Switching Between Opiates

Oral morphine $\div 2$ $=$ **Oral oxycodone**

Oral morphine $\div 2$ $=$ **Subcutaneous morphine**

Oral morphine $\div 3$ $=$ **Subcutaneous diamorphine**

Oral tramadol $\div 20$ $=$ **IV morphine**

Formulation For End of Life

At home + multiple seizures?

Benzodiazepines

Oral

v.s.

Rectal



Nil by mouth / nausea & vomiting + in pain

Metoclopramide / Ondansetron

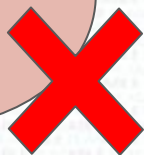
Oral

v.s.

IM

v.s.

IV



Side Effects of ACE Inhibitors

1. Hyperkalaemia ($\uparrow K^+$)

2. Persistent dry cough

On
methotrexate?

For high blood
pressure?

suspect
! Pulmonary fibrosis !

Change to ARB
e.g. Losartan

Symptoms:

1. Dry Cough
2. Dyspnea on mild exertion
3. Diffuse Bilateral Interstitial infiltrates on Chest X-ray

Side Effects of Metoclopramide

Extrapyramidal	Neuroleptic malignant syndrome
Dystonia - <i>‘involuntary movement’</i>	High fever / sweating
Akathisia - <i>‘restlessness’</i>	Muscle rigidity
Bradykinesia - <i>‘slow movement’</i>	Tachycardia
Parkinsonism	Agitation
Tremors	Confusion

Side Effects of Calcium Channel Blocker

3 classes of calcium channel blocker

Diltiazem

Dihydropyridines
E.g. Amlodipine,
Felodipine, Nifedipine

Verapamil

All causes vasodilation therefore side effects:

Flushing, Headache,
***Ankle Swelling, *Gingival Hyperplasia**

↓ Cardiac
output
↓ Heart
rate



↓ Cardiac
output
↓ Heart
rate

Side Effect of Diuretics

Types of diuretic		
Loop Diuretic: Furosemide	Thiazide-like: Indapamide Bendroflume thiazide	Potassium sparing: Spironolactone Eplereone
Hyponatremia	Hyponatremia	Hyponatremia
Hypokalaemia	Hypokalaemia	Hyperkalaemia
Gout	Gout	Gynaecomastia
	Hyperglycaemia	
	Postural Hypotension	

Example:

Bendroflumethiazide → Hyperglycaemia

→  → **Diabetics**

Which diuretic

Brain trainer:

A patient presents with signs and symptoms of cirrhosis, including ascites. What medication should be started?

→ **Spironolactone**

Side Effect of Diuretics

Brain trainer:

A patient on antihypertensive therapy has a fasting blood glucose is 14 mmol/l. What medication could have caused this?

→ **Bendroflumethiazide**

Side Effect of Diuretics

Brain trainer:

A patient with heart failure is started on furosemide. What electrolyte abnormality may occur?

➔ Low sodium, Low potassium

Medications To Be Stopped in Diarrhoea & Vomiting

Diarrhoea + Vomiting = Risk of dehydration

= stop **DAMN** =

D

A

M

N

Diuretics

e.g.

Furosemide,
Bumetanide

↑ **Urinary out**
=

↑ **Dehydration**

ACE-inhibitor

e.g. Ramipril
and **ARB** e.g.
Losartan

↑ **Acute
kidney injury
(AKI)**

Metformin

↑ **Lactic
acidosis in
patient with
↓ Hydration**

NSAID

(Non-steroidal
anti-inflammatory
drugs)

↑ **Acute kidney
injury (AKI)**

Treatment For Alcohol Dependent

AlCohoL - DependEnT

Acamprosate

Reduce and
prevent craving

Disulfiram

Deterrents
(abstinence)

Lorazepam
or
Diazepam

- Hallucination
(oral)
- Seizures
(IV or rectal)

Chlordiazepoxide

Alleviate
withdrawal
symptoms

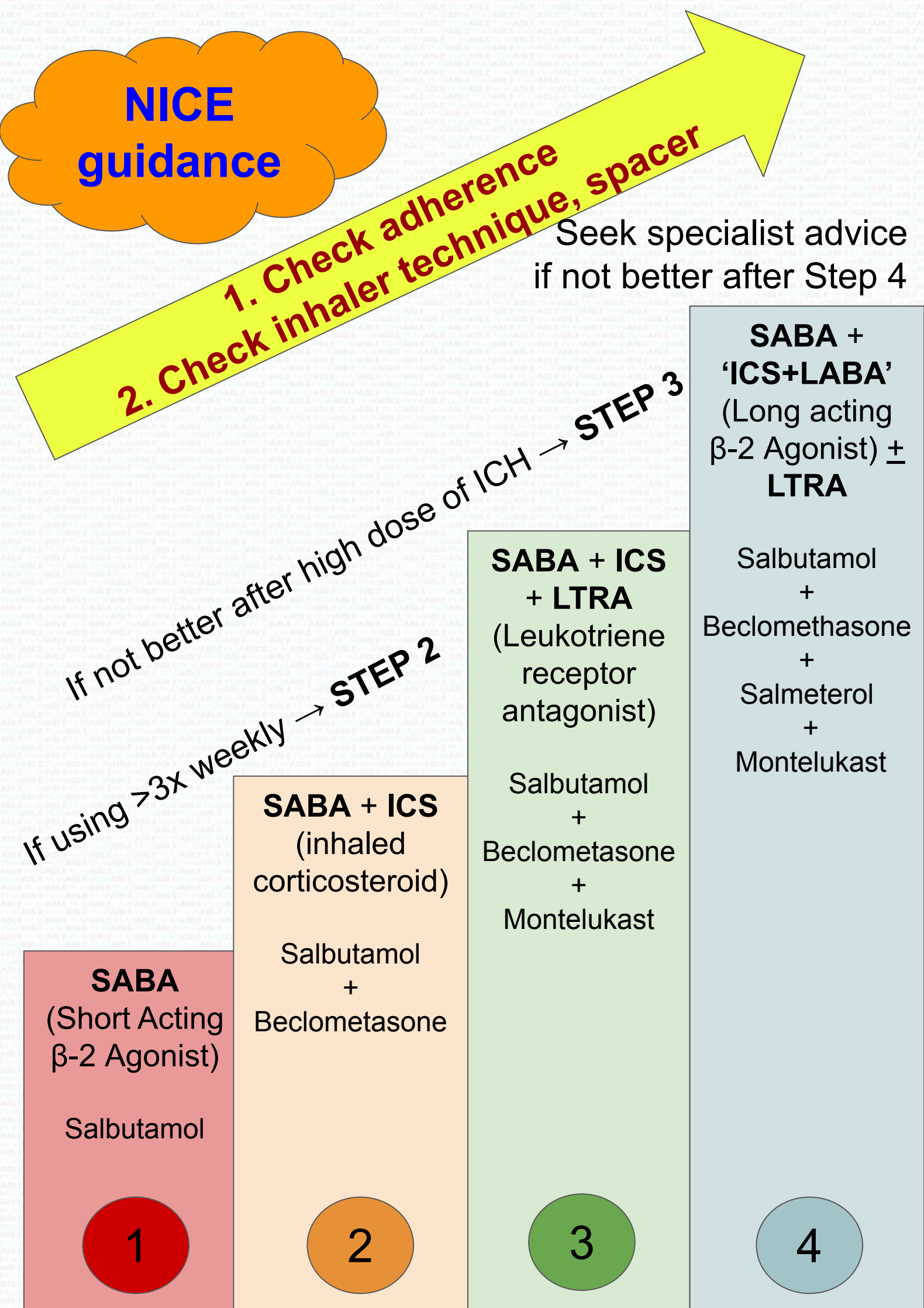
Thiamine or
Vitamin B
complex

Wernicke's
encephalopathy

Enzyme Inducers & Inhibitors

Enzyme inducers = Drugs bye-Bye (↓ Effect)	Enzyme inhibitors = Drug stay-stay (↑ Effect)
🤬CRAP GPs🤬	🤧SICK FACES.COM🤧
Carbamazepine	Sodium Valproate
Rifampicin	Isoniazid
Alcohol - CHRONIC	Cimetidine
Phenytoin	Ketoconazole
Griseofulvin	Fluconazole
Phenobarbital	Alcohol - ACUTE
Sulphonylureas eg. Gliclazide	Chloramphenicol
Remember interaction with combined contraceptive pill and warfarin	Erythromycin - (clarithromycin azithromycin)
	Sulfonamides
	Ciprofloxacin Omeprazole Metronidazole

Asthma Management



L-CHOP for Nausea & Vomiting

Levomepromazine

N&V associated with Parkinson's' diseases
NEVER use
Haloperidol

Cyclizine

1. Increased ICP
2. Hyperemesis gravidarum (1st line)

Haloperidol

1. Renal Failure
2. Hypercalcaemia
3. Drug or Toxin induced

Ondansetron

1. Chemotherapy
2. Radiotherapy
3. Post-operative N&V

Buccal
Prochlorperazine

1. Vertigo
2. Meniere's
3. BPPV
4. Vestibular Neuritis

Haloperidol side effects: sexual dysfunction, gynaecomastia, neuroleptic malignant syndrome

Mechanism of Actions

Tranexamic acid

Inhibit
fibrinolysis

Menorrhagia



N-Acetylcysteine

Protection from
free radicals

**Paracetamol
overdose**



Low molecular
weight heparin

Inhibit
prothrombin to
thrombin

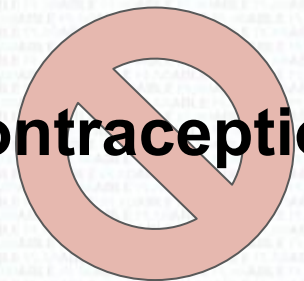
Thrombosis



Copper IUD

Inhibit
implantation

Contraception



Progesterone
only pill
(Levonorgestrel)

Inhibit ovulation

Contraception



Ulipristal acetate
(EllaOne)

Inhibit or delay
ovulation

Contraception



Hair loss

Brain trainer:

A trans-woman (male→ female) is taking multiple medications and as a result has hair loss. What medication could have caused this?

→ Oestrogen

Alcohol Unit Calculations

Golden Formula

$$\text{Strength (\%)} \times \text{Volume (mL)} \div 1000 = \text{.....units}$$

Wine (12%)



Small Glass
125ml = 1.5units



Medium Glass
175ml = 2 units



Large Glass
250ml = 3 units



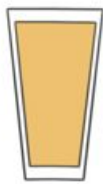
Recommended weekly limits
14 units, at least 2 alcohol free days weekly

Units = (volume [ml] x %ABV) / 1000

Beer (3.5%)



Half pint = 1 unit



1 pint = 2 units

Premium Beer/Lager (5-6%)



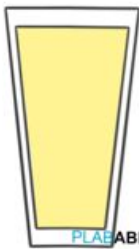
1 pint = 3 units

Spirits (40%)



25ml measure = 1 unit

Cider - often strong!



1 pint = 3 units

Units of Alcohol

UK guidelines recommends:

No more than **14 units per week**

Less than **3 units a day**

At least **2 alcohol-free days per week**

Selective Serotonin Reuptake Inhibitor

1st line for treating depression

Fluoxetine

or

Citalopram

or

Sertraline

Particularly for children and adolescents

Safest option for post myocardial infarction

Duration of actions:

Takes **2 - 4 weeks** for effect to be seen

If NO Response:

2 - 4 weeks

Check patient's adherence

Compliance addressed

After 4 week

↑ Dose or Switch to alternative

Side Effects

1

Gastrointestinal symptoms

Increase risk of GI bleed.

Avoid NSAID, Aspirin, Warfarin, Triptans

2

Hyponatraemia
(Particularly in elderly)

3

Anorgasmia
(Delayed ejaculation)

SSRI

Brain trainer:

Name a cardioselective beta-blocker which can cause bronchoconstriction?

➔ **Atenolol**

Neuroleptic Malignant Syndrome

Causative drugs:

Haloperidol, Metoclopramide

Seen after newly starting a drug or increasing dosage

S T A M P - C

Features:

- **S**weating
- **T**achycardia
- **A**gitation
- **M**uscle Rigidity e.g. neck stiffness
- **P**yrexia (fever)
- **C**onfusion/altered consciousness

- Raised creatinine kinase
- Acute kidney injury (secondary to rhabdomyolysis)
- Leukocytosis

Neuroleptic Malignant Syndrome

Management:

- **STOP** antipsychotic
- Rapid cooling
- IV fluids (prevent renal failure)
- **Dantrolene** - postsynaptic muscle relaxants
- Dopaminergic agent e.g. **Bromocriptine**

Patients with Parkinson's

1.  DO NOT USE metoclopramide with nausea & vomiting
2.  DO NOT USE haloperidol with psychotic symptoms

Management of Acute Psychosis in Parkinson's

1st line : **Quetiapine**

2nd line : **Clozapine**

Points:

1. Typical antipsychotics = contraindicated
e.g. Haloperidol particularly in psychosis and dementia
2. Some antipsychotic can exacerbate PD symptoms
e.g. Olanzapine, risperidone

Patients with **NO** Dementia, Alzheimer or Parkinson's

Haloperidol or olanzapine can be considered as short-term therapy if non-pharmaceutical techniques have failed.

Parkinson's disease

Brain trainer:

What is a known known adverse effect of the sudden cessation of co-careldopa?

→ **Acute akinesia**

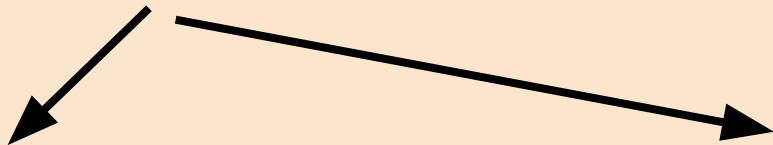
Polypharmacy

- Age ≥ 60
- Taking ≥ 5 medications
- **Common presenting symptoms:**
 - Dizziness
 - Confusion
 - Frequent falls (feeling dizzy moment before fall)
 - Postural hypotension (caused by drugs e.g. antihypertensive, anticholinergic)
- **Watch out for electrolyte imbalance in elderly**
 - Thiazide diuretics →→ Hyponatraemia
 - ACE inhibitors →→ Hyperkalaemia



Lithium For Bipolar Disorder

Before treatment



```
graph TD; A[Before treatment] --> B[Blood check]; A --> C[Physical check];
```

Blood check

1. Liver function test
2. Thyroid function test
3. Renal function test
4. Baseline ECG
5. Full blood count

Physical check

1. Blood pressure
2. Pulse
3. Pregnancy test

Amiodarone for Atrial Fibrillation

Before treatment

Blood check

1. Liver function test
2. Thyroid function test
3. Renal function test
4. Chest x-ray

Physical check

1. Blood pressure
2. Pulse (rate and rhythm)

Fentanyl Patches Or Not?

YES

- When oral route is not possible

i.e. nausea & vomiting

- If there is swallowing difficulties
→ Pain is controlled or **stable**

NO

- Pain is uncontrolled or **unstable**

∴ Fentanyl is slow acting + long half-life

Takes 12-24 hours to achieve therapeutic effect

- Stepping up from controlled pain

i.e. pain stabilized by oral codeine

Low potency —————> High potency

Codeine → Buprenorphine → Morphine → Fentanyl → Oxycodone

Slow acting —————> Fast acting

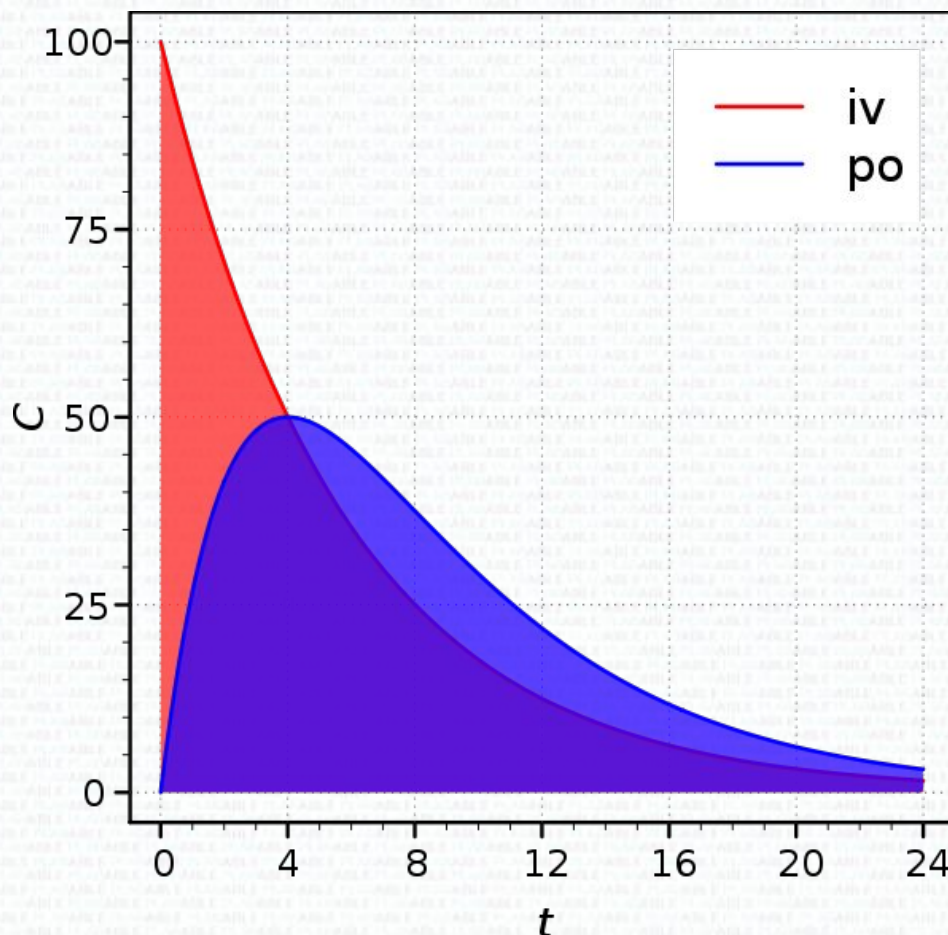
Oral

Transdermal

Intramuscular

Intravenous

Bioavailability



IV drug → Blood stream = Fast acting

High Concentration → Renal excretion



Fast Elimination

Oral → Gut → Blood stream = Slow acting



Hepatic First pass Metabolism



Low concentration

Dose modification

Brain trainer:

A patient with liver failure is taking a lipid soluble medication which strongly binds to albumin and undergoes hepatic first pass metabolism. What dose modification should you make?

→ **Reduce the dose and increase the dosing interval**

Deep Vein Thrombosis



UPS CCTV



Features:

- **Unilateral** - can be bilateral (rare)
- **Pain**
- **Swelling**
- **Changes to skin colour**
- **Commute (Traveling history)**
- **Temperature**
- **Vein distension**

Management:

START - DOAC Or Low molecular weight heparin (s/c)

Until DVT is ruled out

Neuropathic Pain

Causes:

Post-operation, diabetic, post injury, cancer

Features:

Burning

Paresthesia

Itching

Numbness

Ingling

Shooting/Stabbing

Treatment:

1. Amitriptyline
2. Gabapentin
3. Duloxetine
4. Pregabalin



Good
glycaemic
control

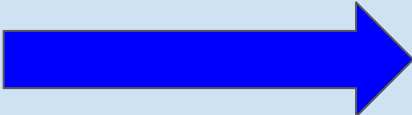


Side Effects of β -adrenoreceptors

β -Blocker  **Bronchospasm**

e.g. Wheezing, SOB, heavy chest

 **Asthma** 

β -Agonist  **Tachycardia**

e.g. Palpitation, muscle twitching, tremors

 **Severe cardiac diseases** 

! REMEMBER !

Both salmeterol (long acting) and salbutamol (short acting) are β -agonists, both have same side effects

Side Effect of Beta-blocker

Brain trainer:

Name a cardioselective beta-blocker which can cause bronchoconstriction?

→ **Atenolol**

Side Effects of β -adrenoreceptors

Salbutamol are used on when required basis
↓
Higher risk of overused

**Higher risk of side effect!
e.g palpitation,
tachycardia**

NSAIDs and aspirin can also aggravates asthma

Increasing theophylline

Medications like theophylline sometimes need to be increased according to certain percentages depending on serum levels.

If you are advised to increase 25% of the dose of medication, this becomes more of a calculation question.

Eg. Please increase theophylline 125 mg by 25%

The calculations would then be:
 $125\text{mg} + 31.25\text{ mg} = 156.25\text{ mg}$

Infusion rates

Brain trainer:

A 40 year old man is having status epilepticus. The decision to start the patient on phenytoin is made. The patient weighs 50 kg and requires 20 mg/kg of phenytoin at a rate of 50 mg/min. Which of the following is the duration of time over which phenytoin must be administered?

→ 50 kg x 20 mg/kg / 50 mg/min = 20 minutes

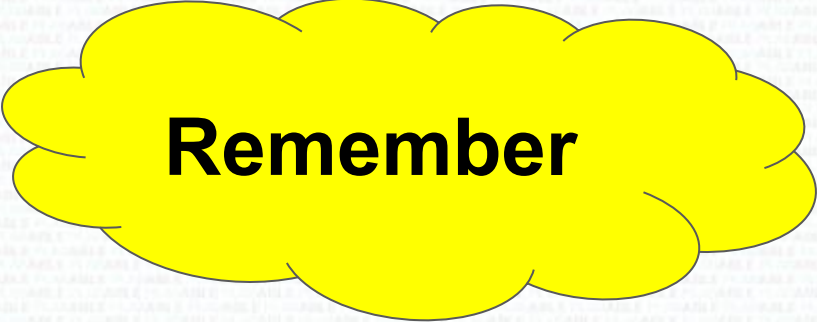
Weight x dose per weight / infusion rate = Time

Opiate Intoxication

Treatment = **Naloxone**

*Naloxone has a **short plasma half-life***

~ 1 to 1.5 hours



Remember

Half-life of overdosed opiates

e.g. Methadone



> GREATER THAN >

Half-life of naloxone



**Naloxone needs
REPEATED doses.
i.e. more than once daily**

Opiate Addiction

Methadone is 1st line for *detoxification*

Mechanism

Mimic opioid properties, reduces withdrawal symptoms of opioid addicts

Opioid Overdose

Brain trainer:

A child presents with symptoms of opioid overdose after accidentally taking methadone. naloxone is administered and the symptoms abate but return a short while later. What caused the symptoms to return?

➔ **Naloxone half-life is shorter than methadone**

Heroin Withdrawal

Timeframe:

Start 12 hrs after last dose

Peak at 24-48 hrs after

Features:

AAS...IIF....

Abdominal pain / Arthralgia

Agitation

Sweating

Increase body secretion (diarrhoea, tears)

Insomnia

Flu-like symptoms

Heroin Overdose/ Intoxication



Low respiratory rate (**Respiratory depression**)

Low blood pressure

Low heart rate

Low gastrointestinal motion (**Constipation**)

Reduced pupil size (**Pinpoint pupils**)

Give Naloxone

Cocaine Withdrawal

Timeframe:

Within few hours after last dose

Peak in a few days

Features

M **I** **N** **D**

Muscle aches

Irritability

Insomnia

Depression

Cocaine Overdose/ Intoxication



- High respiratory rate
- High blood pressure
- High heart rate
- Increased pupils size
(Mydriasis)
- Hyperthermia and sweating
- Restlessness
- Agitation

Give IV Diazepam

Alcohol Withdrawal

Timeframe:

Within 6-12 hours after last dose

Features:

Sweating

Tremor

Anxiety

Tachycardia

Seizures (at 36 hours)

Delirium tremens (at 48-72 hours)

- Coarse tremor, confusion, delusions, auditory and visual hallucinations, fever, tachycardia

STATS.....D

Management:

First Line - **Benzodiazepines**

e.g. Chlordiazepoxide or lorazepam in hepatic failure

Second Line - **Carbamazepines**

Phenytoin is not as effective in treatment of alcohol withdrawal seizures

Benzodiazepines Withdrawal

Timeframe:

Within 1-4 days after last dose
Peak at 2 weeks

Features:

Panic Attack

Agitation

Insomnia

ANxiety

PAIN

Benzodiazepines are used to treat panic attack and anxiety

ANXIETY



Ecstasy Overdose/ Intoxication

Features:

- **Neurological:**
→ Agitation, anxiety, confusion, ataxia
- **Cardiovascular:**
→ Tachycardia, hypertension
- **Hyponatraemia**
- **Hyperthermia**
- **Rhabdomyolysis**
- **Uncontrolled body movement, Trismus**

Management:

1. Supportive
2. Dantrolene - for hyperthermia if simple measures fail

LSD overdose / intoxication

(Lysergic acid diethylamide)

Features:

- Mydriasis
- Flushing
- Sweating
- Hyperreflexia
- Diarrhoea
- Paraesthesia
- Delusions
- Hallucination (Pathognomonic)

**A patient seeing sounds and
smelling colours**

Respiratory Infections

Infection	Antibiotic
Community acquired pneumonia (Mild)	Amoxicillin
Community acquired pneumonia (Moderate)	Amoxicillin + Clarithromycin
Community acquired pneumonia (Severe)	Co-amoxiclav (amoxicillin + clavulanic acid) + Clarithromycin
Pneumocystis jirovecii (<i>P. Carinii</i>)	Co-Trimoxazole (trimethoprim + sulfamethoxazole)
Tuberculosis (<i>TB</i>)	1st 2 months - RIPE Rifampicin, Isoniazid, Pyrazinamide, Ethambutol Next 4 months - RI Rifampicin, Isoniazid
Aspiration pneumonia (community acquired)	Amoxicillin + Metronidazole

CNS Infection

Infection	Antibiotic
In-hospital meningitis	Cefotaxime IV or Ceftriaxone IV
Out-of-hospital meningitis	Benzylpenicillin
Listeria meningitis	Amoxicillin or Ampicillin + Gentamicin
Cryptococcal meningitis	Amphotericin B
Meningitis prophylaxis (for contacts)	1st Line: Ciprofloxacin 2nd Line: Rifampicin

Genitourinary Infection

Infection	Antibiotic
Lower uncomplicated UTI	Trimethoprim or Nitrofurantoin
<i>Candida albicans</i> (Vulvovaginal Candidiasis)	Clotrimazole or Fluconazole
<i>Trichomonas vaginalis</i> (Bacterial vaginalis)	Metronidazole
Cervicitis (Chlamydia)	1st line - Doxycycline 2nd line or pregnancy - Azithromycin
Cervicitis (Gonorrhoea)	Ceftriaxone IM or Clprofloxacin (if sensitivity is known)
Pelvic inflammatory diseases	1st line - Ceftriaxone + Metronidazole + Doxycycline 2nd line: - Metronidazole + (Ofloxacin OR Moxifloxacin)
Syphilis	Penicillin G
Genital herpes (HSV)	Aciclovir

Ringworm

Brain trainer:

What treatment is indicated for a ringworm infection (dermatophytosis)?

→ Clotrimazole cream

Gastrointestinal Infection

Infection	Antibiotics
Salmonella / Shigella / Campylobacter	Erythromycin OR Azithromycin OR Clarithromycin OR Ciprofloxacin
<i>Clostridium difficile</i> (Pseudomembranous colitis)	For mild: Oral Metronidazole For severe: Vancomycin
<i>Helicobacter pylori</i>	OAC or OAM Omeprazole Amoxicillin Clarithromycin or Metronidazole



Ear, Nose & Throat Infection

Infection	Antibiotics
Acute otitis media (Bacterial)	Amoxicillin
URTI (Pharyngitis, Tonsillitis, Laryngitis)	Phenoxymethylpenicillin

Other Infections

Infection	Antibiotics
Cellulitis Mastitis Diabetic Foot infection	Flucloxacillin
Septic arthritis Osteomyelitis	Flucloxacillin ± IV Gentamicin or Metronidazole
Scabies	Permethrin
Toxoplasmosis	Pyrimethamine + Sulfadiazine

Antibiotic Interaction

Macrolides
e.g.
Clarithromycin

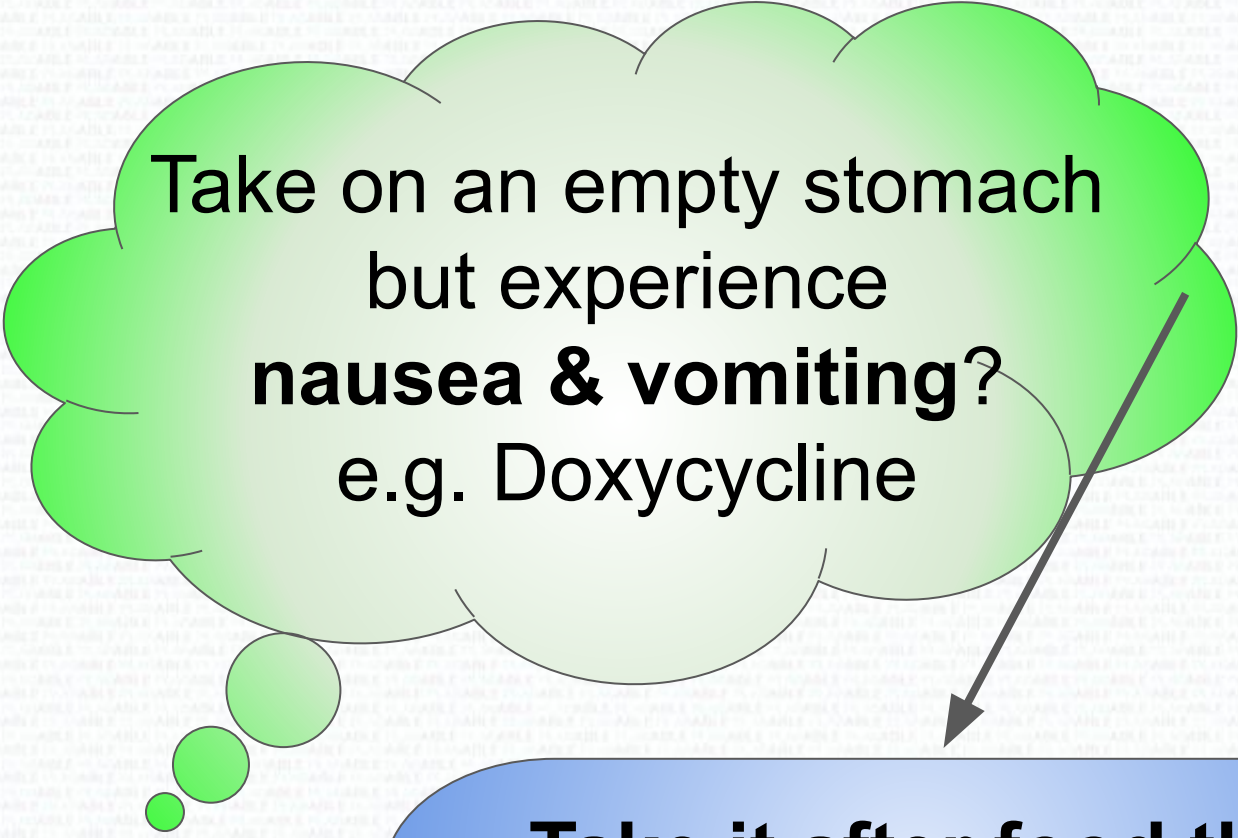


Statin



Rhabdomyolysis

Antibiotic Side Effects



Take on an empty stomach
but experience
nausea & vomiting?
e.g. Doxycycline



Take it after food then

Medication Red Flags

Bisphosphonates?
→ GI bleed

Heartburn
Chest Pain

Headache

Warfarin?
→ Subdural Haematoma

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